

Week 3 Progress Report: AI Agent Automation Proposals

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Department: AI & ML Department

1. Executive Summary

During **Week 3** of the **SafeX Solutions Internship**, the **AI & ML Department** focused on designing **AI Agent Automation Proposals** tailored to solve specific operational bottlenecks for various target businesses.

Our group established a shared modular framework, allowing independent developers to build production-grade modules. My specific deliverable was to build the **AI Social Media Scheduler & Caption Generator** designed to streamline marketing operations for online tutoring platforms like **LearnHub Academy**. The code, unit tests, and comprehensive case studies are completed and verified.

2. Weekly Achievements & Deliverables

Technical Deliverables Completed:

- **`engine.py` (NLP & Context Planner Engine):** Custom generator rules mapping content topics (e.g. coding basics, AI utilities, spreadsheets) to specific platform standards (LinkedIn, Instagram, X/Twitter, Facebook).
- **`ui.py` (Interactive Streamlit Dashboard):** Features dynamic day-by-day tab editors, real-time character boundary audits, Midjourney visual prompts, and multi-format downloads (CSV & JSON).

- ``test_social_media_scheduler.py`` (**Automated Verification Suite**): 5 unit test cases testing calendar structure, character limits constraints (X/Twitter 280 limit), tone variations, and image prompts.
- ``social_media_scheduler.md`` (**Individual Case Study**): Full report detailing business objectives, system workflows, and testing benchmarks.

3. Verification & Testing Metrics

- **Unit Tests Run:** 12 tests passed successfully.
- **Success Rate:** 100%
- **Performance:** Local simulated AI generator execution latency is under 5ms, avoiding external LLM API usage costs while maintaining structural consistency.
- **Compliance Rate:** 100% character audit accuracy, successfully identifying and warning users when text lengths exceed target platform limits.

4. Next Steps & week Plans (Week 4)

1. **API Integration Phase:** Expand the scheduler from simulated integration proposals to active staging integration using sandbox API keys.
2. **Database Persistence:** Connect the interactive calendar edits to a local SQLite/PostgreSQL instance to preserve state across application reruns.
3. **Collaboration Sync:** Support other group members in upgrading their scaffolding modules to submission-ready status.